Conservation versus Trajectory Accuracy in Direct Gravitational N-Body Integration: A Benchmark of Euler, Velocity-Verlet, RK4, and Yoshida4

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Abstract

Numerical integration of gravitational N-body systems requires balancing trajectory accuracy, conservation, and computational cost. This study benchmarks direct Newtonian integration using forward Euler, velocity-Verlet, classical fourth-order Runge–Kutta (RK4), and Yoshida4. Experiments include analytical two-body validation, timestep convergence, long-duration eccentric motion, multi-year planetary configurations, binary dynamics, conservation diagnostics, direct-force scaling, and gravitational softening. For an eccentricity-0.9 Kepler orbit, an analytical solution of Kepler's equation provides the reference for fixed-step methods. The measured global position-error orders over the finest usable grid levels are 1.93 for velocity-Verlet, 4.05 for RK4, and 4.00 for Yoshida4. Over 1,000 orbital periods, the long-duration resolution sweep (Table 3) shows that velocity-Verlet has the largest final phase error at all three tested resolutions (5,000, 10,000, and 20,000 steps per orbit), while Yoshida4 has the smallest. At 10,000 steps per orbit, velocity-Verlet and Yoshida4 maintain bounded energy error despite substantial phase separation; RK4 has lower phase error than velocity-Verlet at this resolution but exhibits secular energy drift. A one-period figure-eight return test approaches a roughly 5×10^{-8} discrepancy floor associated with rounded initial conditions. The Pythagorean close-encounter test is not used to rank methods because the tested integration is under-resolved. The results illustrate that trajectory accuracy and invariant preservation are distinct criteria; they do not establish a new algorithm or superiority over established N-body packages.

Keywords: computational astrophysics; gravitational N-body problem; numerical integration; symplectic methods; conservation laws; work–precision; orbital dynamics

1. Introduction

The gravitational N-body problem describes the coupled motion of masses interacting through gravity. Numerical solutions support studies of planetary motion, asteroid and comet propagation, satellite trajectories, binary stars, and stellar systems. Even for Newtonian forces, the computed trajectory depends on the integration scheme, timestep, initial conditions, arithmetic precision, and diagnostic selected. A visually plausible orbit is not, by itself, evidence of numerical accuracy.

Direct summation evaluates every pairwise interaction and has $O(N^2)$ force-evaluation cost (Harfst et al., 2007; Aarseth, 2003). Its transparency makes it useful for small systems and validation studies, although the cost becomes prohibitive for large particle counts. Numerical integrators introduce a separate trade-off. Forward Euler is simple but generally poor for long conservative trajectories. Velocity-Verlet is second-order, time-reversible, and symplectic for the separable Hamiltonian systems considered here. Classical RK4 is not symplectic but can deliver high finite-time trajectory accuracy. Yoshida4 composes second-order symplectic maps to obtain fourth-order accuracy. Adaptive DOP853 provides a high-accuracy numerical baseline (Hairer, Nørsett, & Wanner, 1993), while an analytical Kepler solution offers a stronger reference for the two-body problem.

Energy error, angular-momentum error, and phase or position error are different quantities. A method may preserve energy within a bounded envelope yet accumulate phase error, and a close encounter can amplify small state discrepancies. This paper therefore combines analytical validation, convergence studies, invariant diagnostics, long-duration eccentric motion, work–precision analysis, and limited multi-body tests. The contribution is an inspectable benchmark for computational-physics education and method interpretation—not a new integration algorithm. External REBOUND and JPL Horizons comparisons are not claimed. The central questions are whether observed convergence orders match formal method orders, whether long-time energy behaviour tracks phase accuracy, and

how much fixed-step accuracy is obtained per force evaluation. The experiments are chosen to answer these questions with analytical references where available, while explicitly separating finite-time accuracy from long-time geometric behaviour.

2. Related work

Geometric numerical integration explains why symplectic methods can preserve qualitative features of Hamiltonian systems over long integrations (Hairer, Lubich, & Wanner, 2006). Yoshida (1990) developed composition techniques for higher-order symplectic methods, while Wisdom and Holman (1991) introduced symplectic mappings tailored to long-term planetary dynamics. REBOUND is an open-source N-body framework (Rein & Liu, 2012); WHFast and IAS15 provide established symplectic and adaptive high-order alternatives (Rein & Tamayo, 2015; Rein & Spiegel, 2015). Brouwer's (1937) analysis is relevant to the interpretation of long-run numerical error. Barnes and Hut (1986) introduced a hierarchical tree force approximation to reduce large-N cost. The present direct-force implementation is not intended to match the maturity, features, or performance of these packages.

3. Mathematical model

For body i with mass m_i, position r_i, and velocity v_i, the unsoftened Newtonian equations are

$$d^2r_i/dt^2 = G \sum_{j \neq i} m_j (r_j - r_i) / |r_j - r_i|^3$$

The total energy, angular momentum, linear momentum, and centre of mass are

$$E = \sum_i \tfrac{1}{2} m_i |v_i|^2 - \sum_{i<j} G m_i m_j / |r_i - r_j|$$

$$L = \sum_i m_i (r_i \times v_i) \; ; \; P = \sum_i m_i v_i \; ; \; r_cm = (\sum_i m_i r_i)/(\sum_i m_i)$$

Relative energy error is $\delta E(t) = |E(t) - E(0)| / |E(0)|$. Relative angular-momentum error is defined analogously when the initial angular momentum is nonzero. Position discrepancy is the Euclidean difference from a reference state at the same time, maximized over sampled times and bodies unless noted otherwise. The astronomical unit is exactly 149,597,870,700 m under Resolution B2 (IAU, 2012).

For softening length ε, the pair potential uses $\sqrt{(|r_i - r_j|^2 + \varepsilon^2)}$ in place of separation. Softening changes the physical force law; it is a modelling choice rather than an automatic accuracy improvement.

4. Numerical methods and experimental design

Forward Euler uses the beginning-of-step acceleration. Cached velocity-Verlet advances positions and then velocities using

$$r_{n+1} = r_n + h\, v_n + \tfrac{1}{2} h^2 a_n \; ; \; v_{n+1} = v_n + \tfrac{1}{2} h (a_n + a_{n+1})$$

Classical RK4 applies four stages to the first-order state (r, v). Yoshida4 composes three Verlet maps with coefficients $w_1 = 1/(2 - 2^{(1/3)})$ and $w_0 = -2^{(1/3)}/(2 - 2^{(1/3)})$, using substeps $w_1 h$, $w_0 h$, and $w_1 h$. With acceleration caching, the estimated new acceleration evaluations per macro-step are one for velocity-Verlet, four for RK4, and three for this Yoshida4 composition. DOP853 work is counted using the solver's reported right-hand-side evaluations.

The circular Sun–Earth analogue is compared with its analytical two-body solution. A 1.0–0.8 solar-mass binary provides a second analytical check. Circular convergence uses timesteps of 4, 2, 1, 0.5, 0.25, and 0.125 days. The five-year integrator comparison uses a 2-day step; the ten-year circular radius-stability test uses velocity-Verlet at 0.5 day. The eccentric Kepler problem has semimajor axis a = 1, eccentricity e = 0.9, total mass parameter μ = 1, and periapsis r = 0.1a; the exact orbit is reconstructed by solving $M = E - e \sin E$. The long-duration experiment integrates 1,000 periods with 10,000 fixed steps per period. The figure-eight return test uses the rounded published initial conditions and integrates one period. The Pythagorean problem is retained only as a qualitative close-encounter caution.

Table 1. Principal initial conditions and experiment scope

Problem	Initial conditions / units	Purpose and duration
Circular Sun–Earth	SI units; separation 1 au; barycentric circular	Analytical validation; 1 year, Δt = 0.5 day

	state; relative speed √[G(M_sun + M_earth)/a]	
Eccentric Kepler	Dimensionless μ = 1, a = 1, e = 0.9; periapsis r = 0.1, tangential speed √19	Analytical Kepler reference; 100–100,000 steps/orbit; long run 1,000 orbits at 10,000 steps/orbit
Figure-eight	Equal masses; r_1=(0.97000436,−0.24308753), r_2=−r_1, r_3=(0,0); v_1=v_2=(0.466203685,0.432365730), v_3=−2v_1	One-period RK4 return error; published rounded initial conditions
Pythagorean	Dimensionless G=1; masses (3,4,5), positions (1,3), (−2,−1), (1,−1), initial velocities zero	Qualitative close-encounter example only; not used to rank methods
Sun–Earth–Mars / binary	Direct Newtonian point masses; barycentric states	Perturbation and second analytical validation
Scaling / softening	Direct summation; N scaling and ε sweep	Cost trend and sensitivity; limited exploratory tests

Software environment: Python 3.13.5, NumPy 2.3.5, SciPy 1.17.0, Matplotlib 3.10.8, Numba 0.65.1, and pandas 2.2.3 on Linux x86_64. The recorded CPU model was AMD EPYC 9V74 80-Core Processor; three logical CPUs were available to the process. Timings are exploratory and should not be generalized beyond this environment.

5. Results

5.1. Analytical Sun–Earth validation

At Δt = 0.5 day over one year, velocity-Verlet produced maximum Earth position error 1.619×10^{-4} au, maximum relative energy error 1.368×10^{-9}, maximum relative angular-momentum error 1.635×10^{-15}, and maximum centre-of-mass displacement 2.340×10^{-20} au. This checks the force implementation and diagnostics for a comparatively benign orbit; it does not establish accuracy for arbitrary geometries.

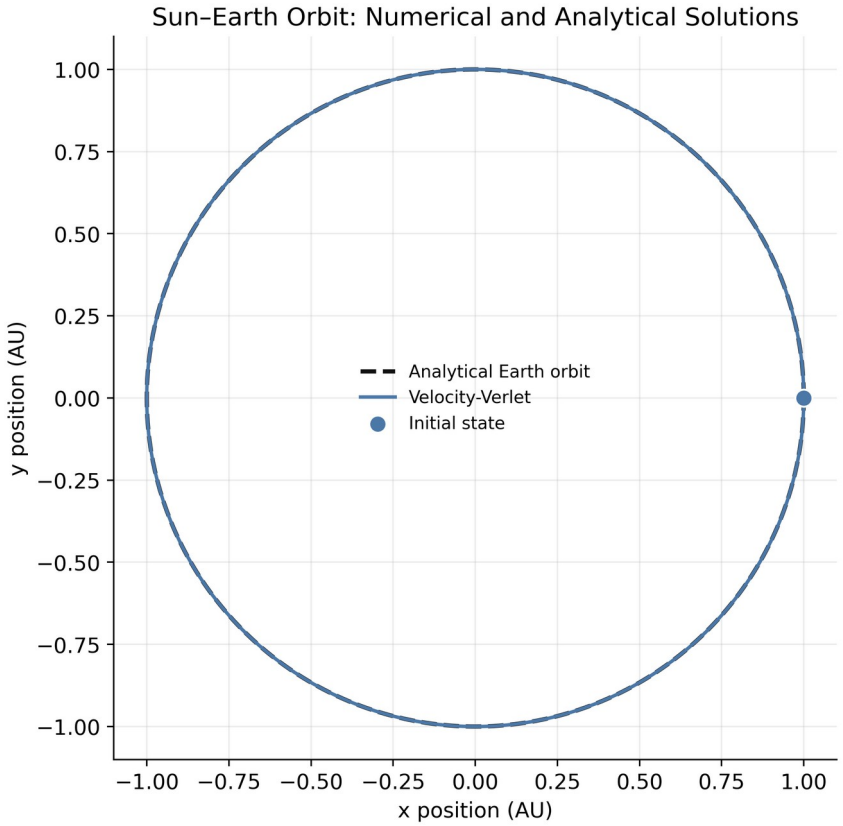


Figure 1. Numerical Sun–Earth orbit compared with the analytical circular two-body solution; spatial axes use equal scaling.

5.2. Five-year integrator comparison

At Δt = 2 days, all four methods were run over the same nominal five-year interval (913 steps; 1,826 days). Table 2 reports maximum errors over each run. RK4 has the smallest maximum Earth-position discrepancy, while Yoshida4

has the smallest energy error, and Verlet and Yoshida4 are both at machine precision for angular momentum. These values are from the five-year experiment; the separate one-year convergence results are not mixed into this comparison.

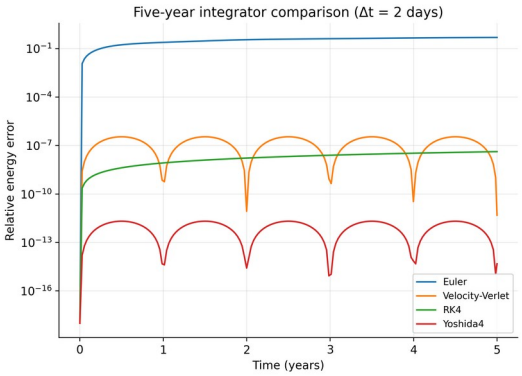


Figure 2. Relative energy error in the multi-year integrator comparison.

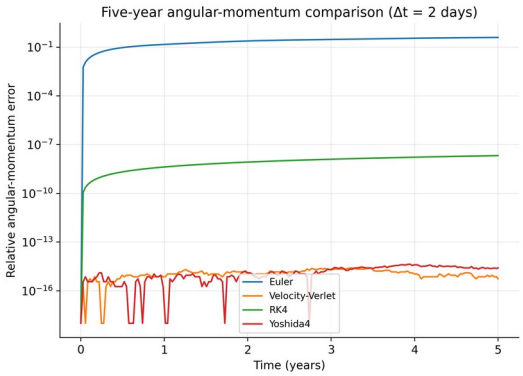


Figure 3. Relative angular-momentum error in the five-year integrator comparison.

Table 2. Five-year nominal comparison at Δt = 2 days (913 steps; 1,826 days).

Method	Max $\lvert\Delta E/E_0\rvert$	Max $\lvert\Delta L/L_0\rvert$	Max Earth position error (au)
Euler	4.935×10^{-1}	3.991×10^{-1}	2.751
Velocity-Verlet	3.497×10^{-7}	2.726×10^{-15}	1.250×10^{-2}
RK4	4.207×10^{-8}	2.104×10^{-8}	1.999×10^{-6}
Yoshida4	2.085×10^{-12}	6.723×10^{-15}	4.082×10^{-5}

5.3. Timestep convergence and the circular-orbit energy slope

The dedicated fine Euler sweep from 10^3 to 10^6 steps per year gives fitted position-error order 0.999 over the four finest points, recovering first-order behaviour after the coarser fit had saturated. The main circular convergence study shows second-order position convergence for velocity-Verlet and fourth-order convergence for RK4 and Yoshida4 before roundoff limits the smallest errors. For velocity-Verlet, the fitted energy-error slopes are 3.32 for the circular orbit and 2.04 for e = 0.9 over the selected fit windows. The eccentric slope is consistent with second-order behaviour. The circular slope is steeper but does not establish a general fourth-order law; a symmetry-related cancellation of a leading error contribution is a plausible, setup-specific interpretation, and the finest circular points approach the floating-point floor. Global trajectory order remains two.

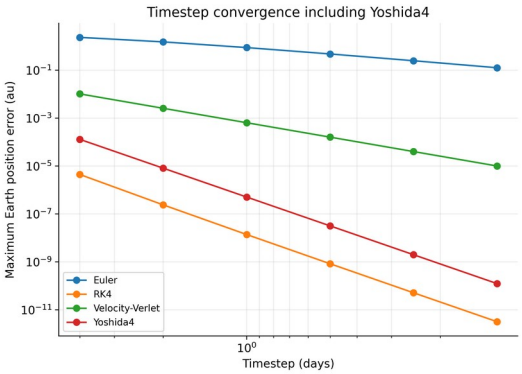


Figure 4. Maximum position error versus timestep for the main methods.

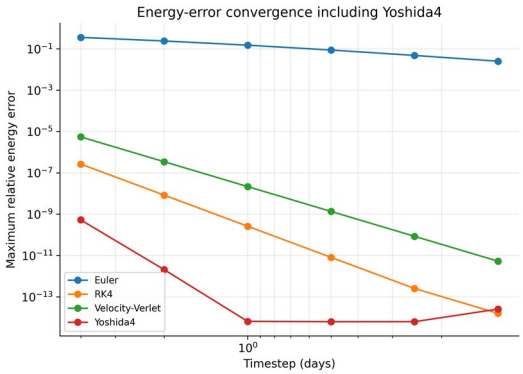


Figure 5. Maximum relative energy error versus timestep; the measured slope depends on quantity and orbit.

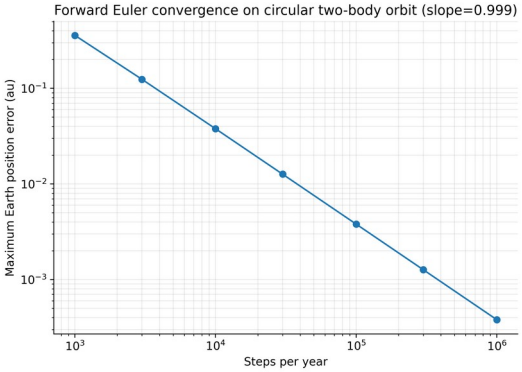


Figure 6. Fine-grid Euler convergence recovers the expected first-order trend.

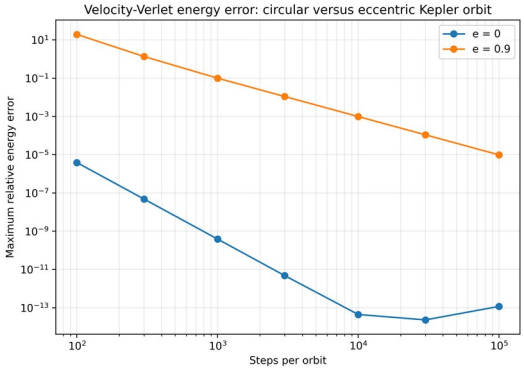


Figure 7. Velocity-Verlet energy-error slopes for circular and eccentric Kepler orbits.

5.4. Long-duration eccentric orbit: conservation versus phase accuracy

To test the central thesis over a longer interval than the one-orbit convergence study, the e = 0.9 Kepler problem was integrated for 1,000 orbital periods using 10,000 fixed steps per period (10 million macro-steps per method). Diagnostics were sampled 200 times per orbit against the analytical Kepler solution. Maximum relative energy error was 9.85×10^{-4} for velocity-Verlet, 3.58×10^{-6} for RK4, and 2.46×10^{-7} for Yoshida4. At the final sample, absolute unwrapped true-longitude discrepancies were approximately 9.44 rad, 0.685 rad, and 0.101 rad, respectively. Table 3 reports the two additional resolutions. Velocity-Verlet's trajectory is effectively lost relative to the analytical orbit over this long integration: its maximum position discrepancy remains $|\Delta r|/a \approx 1.88–2.02$ across all three resolutions, including 20,000 steps per orbit. A discrepancy approaching 2a means that, at some sampled times, the numerical and reference positions are nearly opposite one another—consistent with a phase separation of about half an orbit—even though the energy error remains bounded. Thus, saying only that phase error "accumulates" understates the observed loss of pointwise trajectory agreement. Yoshida4 has the smallest final phase error at each tested resolution, while velocity-Verlet has the largest. For RK4, the final phase error falls by about 3.8× from 5,000 to 10,000 steps per orbit and then by about 29× from 10,000 to 20,000. This nonuniform change is not the clean 16× reduction expected from a simple fourth-order asymptotic phase-error regime under timestep halving; the present sweep therefore does not establish that RK4 has reached that regime. These rankings and scaling observations apply to the tested orbit, duration, diagnostics, and step sizes.

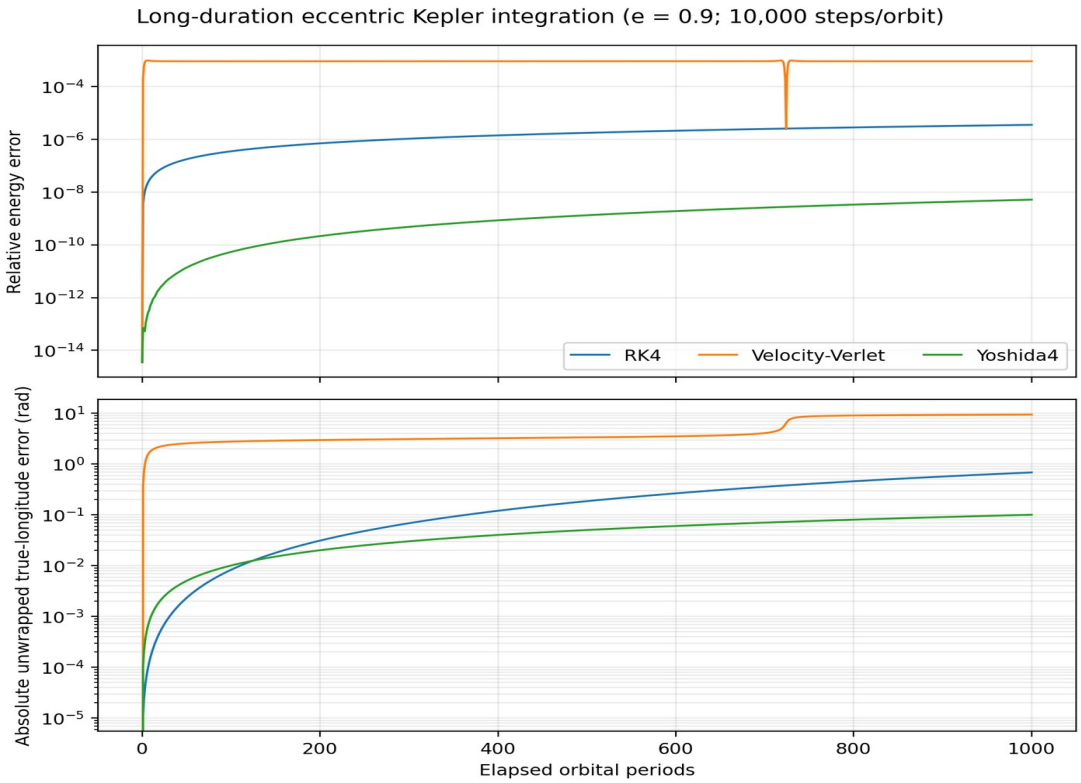


Figure 8. Once-per-orbit display of energy-error magnitude and true-longitude discrepancy for e = 0.9 over 1,000 periods at 10,000 steps/orbit. Tabulated maxima use 200 diagnostic samples per orbit.

The adaptive DOP853 result belongs to the separate one-orbit work–precision experiment and is not included in this long-run fixed-step comparison. Adaptive tolerances, step counts, and phase accumulation would need to be reported explicitly for a comparable long-duration adaptive run.

Table 3. Long-duration resolution sweep; maximum errors use 200 diagnostic samples per orbit.

Steps/orbit	Method	Max $\lvert\Delta E/E_0\rvert$	Max $\lvert\Delta r\rvert/a$	Final $\lvert\Delta v\rvert$ (rad)
5,000	Velocity-Verlet	3.942×10^{-3}	2.015	35.008
5,000	RK4	1.141×10^{-4}	0.9658	2.625
5,000	Yoshida4	3.915×10^{-6}	0.1495	1.228
10,000	Velocity-Verlet	9.850×10^{-4}	2.002	9.441
10,000	RK4	3.576×10^{-6}	0.07208	0.6851
10,000	Yoshida4	2.456×10^{-7}	0.01008	0.1007
20,000	Velocity-Verlet	2.462×10^{-4}	1.880	3.046
20,000	RK4	1.125×10^{-7}	0.002345	0.02344
20,000	Yoshida4	1.536×10^{-8}	0.0006309	0.006309

The sharp velocity-Verlet energy-error dip near 724 periods is not a sudden loss of stability. Figure 8 displays once-per-orbit samples of the absolute, phase-dependent energy error; as the numerical orbit drifts in phase relative to the analytical orbit, one sample passes close to zero, which the logarithmic axis magnifies. The rapid change in true-longitude error is continuous and occurs as the two solutions become differently phased near periapsis, where angular motion is fastest.

5.5. Circular long-term behaviour

In the ten-year circular experiment, position discrepancy grows as phase differences accumulate while the Earth–Sun separation remains close to one au. This reinforces the distinction between qualitative orbital stability and pointwise trajectory agreement. A symplectic method does not eliminate phase error or guarantee high pointwise accuracy at arbitrary step sizes.

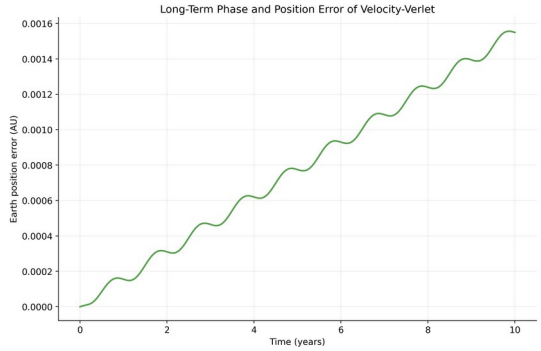


Figure 9. Position error relative to the analytical circular reference over ten years.

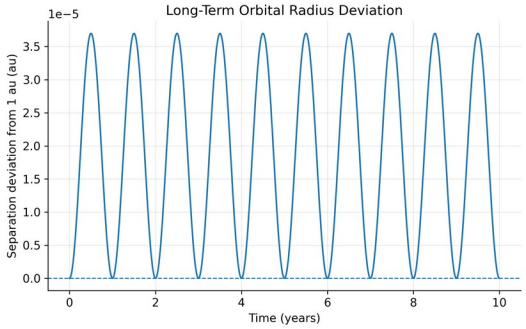


Figure 10. Earth–Sun separation over ten years; bounded radius is not exact phase agreement.

5.6. Third-body perturbation and binary validation

Adding Mars to the Sun–Earth model changed the Earth trajectory through the additional gravitational acceleration. At a 1-day step over five years, the maximum separation between the two Earth trajectories was 2.775×10^{-4} au and the final separation was 2.573×10^{-4} au. These are differences between idealized models, not errors against an observed ephemeris. For a circular binary with masses 1.0 and 0.8 solar masses and a 1-au separation, the maximum body-position discrepancy over five years at a 0.5-day step was 1.075×10^{-3} au.

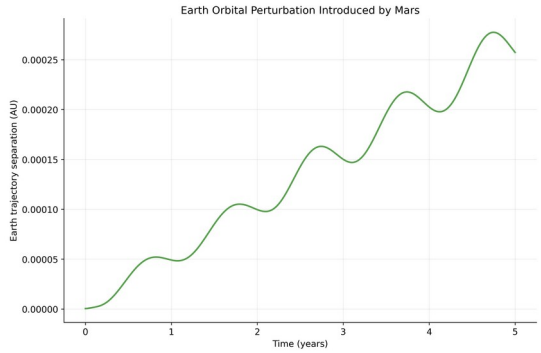


Figure 11. Earth trajectory separation with and without Mars over five years.

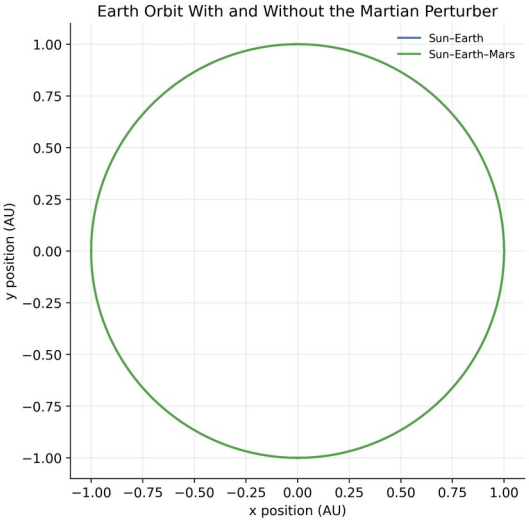


Figure 12. Two-body and three-body Earth trajectories, with equal aspect ratio.

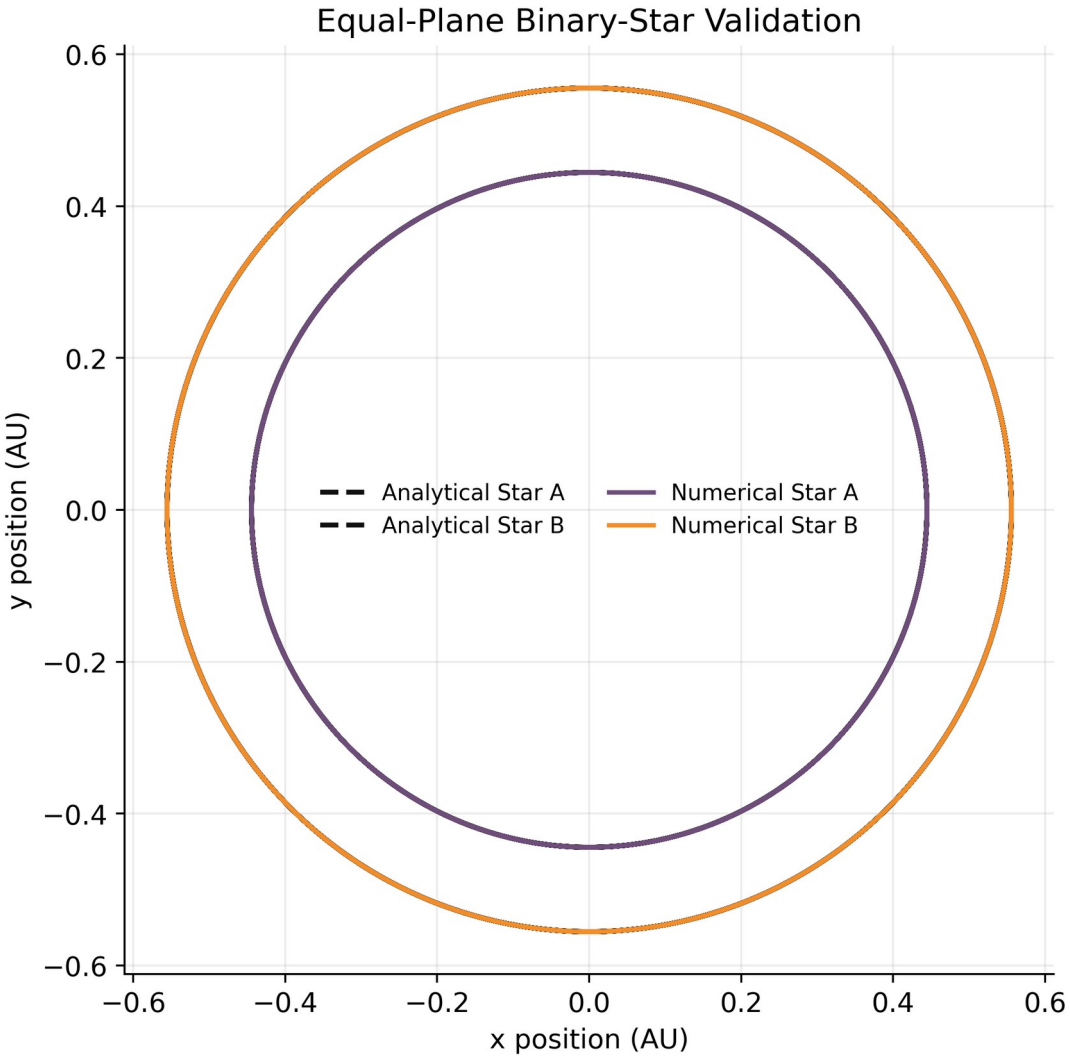


Figure 13. Numerical and analytical trajectories for the circular 1.0–0.8 solar-mass binary; axes use equal aspect ratio.

5.7. Momentum and centre-of-mass diagnostics

For an isolated system, pairwise Newtonian forces cancel in the total momentum balance. The computed centre-of-mass displacement and linear-momentum error remain near floating-point residual levels for the tested Sun–Earth–Mars configuration. These are implementation-level checks; values near roundoff should not be interpreted as meaningful relative accuracy beyond the normalization and precision of the calculation. The figures plot the computed positive values without replacing exact zeros by a plotting floor.

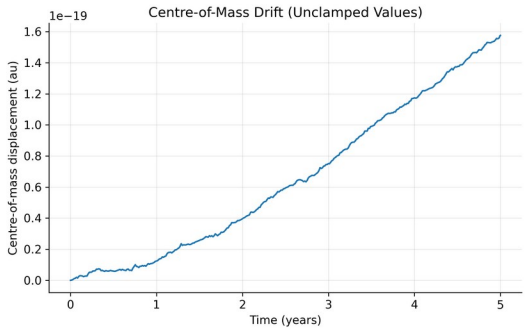


Figure 14. Computed centre-of-mass displacement; positive measured values only.

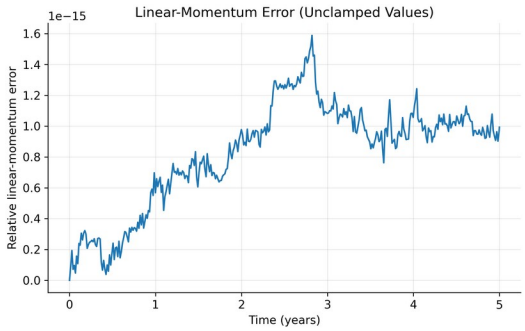


Figure 15. Relative linear-momentum error; positive measured values only.

5.8. Direct-force scaling

The timing protocol uses seven repeats of five velocity-Verlet steps for N = 16, 32, 64, 128, 256, and 512 bodies. The log–log fit is restricted to N ≥ 64 to reduce small-system overhead. The trend is consistent with the $O(N^2)$ cost of direct pairwise summation, but the workload is short and timings depend on implementation, hardware, and system load.

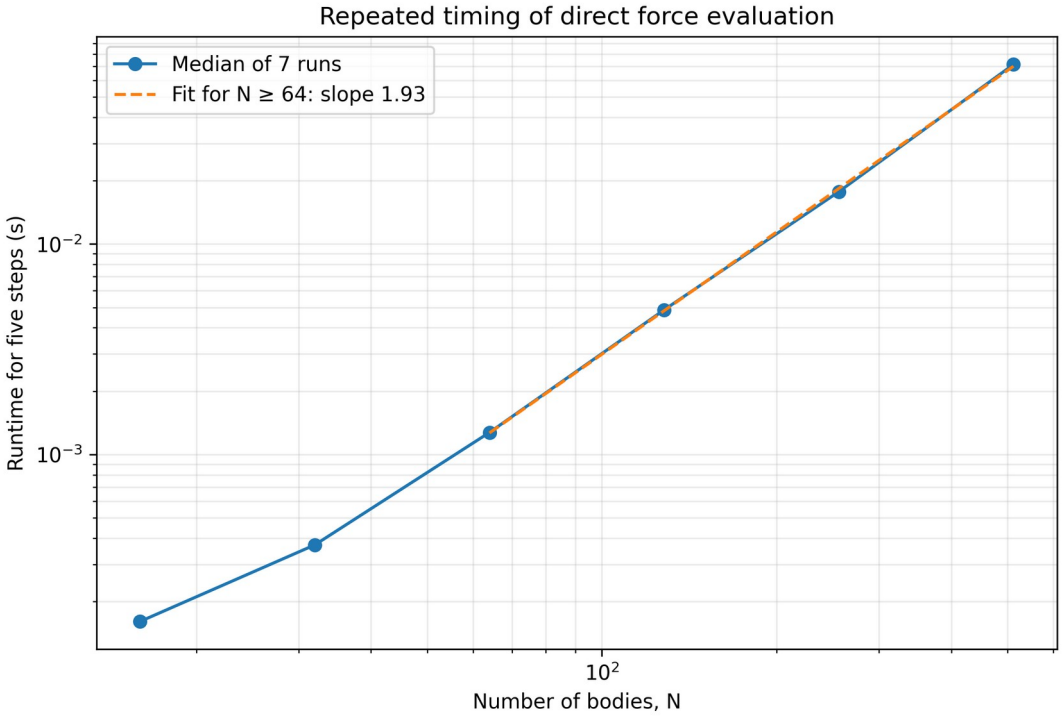


Figure 16. Repeated timing of five direct-force integration steps; points summarize repeated measurements and the fit uses the stated large-N subset.

5.9. Softening and work–precision

In the controlled binary experiment, softening values through ε = 0.001 au changed maximum relative energy error modestly, while ε = 0.005 au increased it to about 1.52×10^{-9}. Because this binary has no close encounter, the result should not be generalized to encounter-dominated systems.

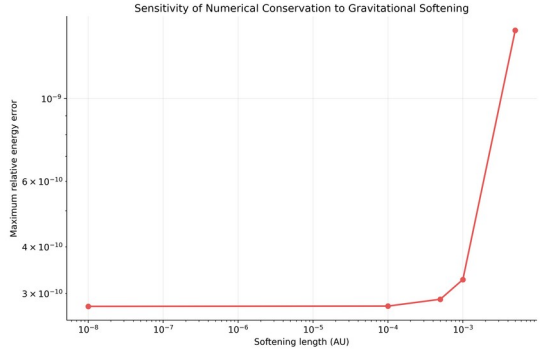


Figure 17. Maximum relative energy error versus softening length; zero softening is shown as the unsoftened reference.

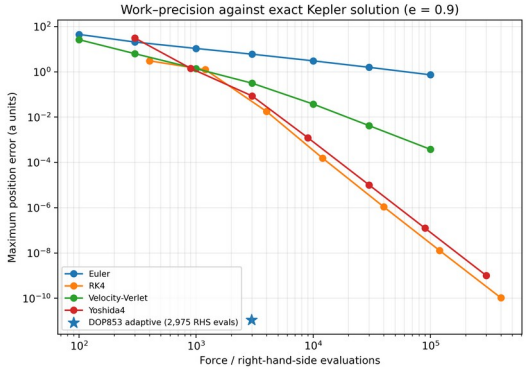


Figure 18. Work–precision comparison against the analytical eccentric Kepler solution; fixed-step work uses cached acceleration evaluation counts, while DOP853 uses the solver-reported evaluation count.

The eccentric work–precision comparison uses the analytical Kepler solution and spans 10^2–10^5 steps per orbit. The adaptive DOP853 run used 2,975 right-hand-side evaluations and reached maximum position discrepancy 1.11×10^{-11}a. It therefore outperforms the displayed fixed-step points by orders of magnitude at the measured point; any discussion of fixed-step cost must include this adaptive baseline. The number of right-hand-side evaluations is not identical to wall-clock time, and DOP853's adaptive overhead differs from fixed-step methods.

5.10. Eccentric convergence and three-body return test

For the eccentricity-0.9 Kepler problem, the analytical reference is obtained by solving Kepler's equation and reconstructing the orbit, avoiding an approximate numerical reference for fixed-step convergence. Over the four finest usable levels, observed global position-error orders were 1.93 for velocity-Verlet, 4.05 for RK4, and 4.00 for Yoshida4. Euler remains outside its asymptotic regime for the eccentric grid and is not assigned an order from those points.

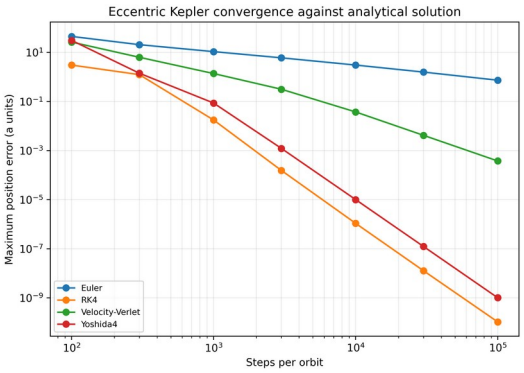


Figure 19. Eccentric-orbit global position-error convergence against the analytical Kepler solution.

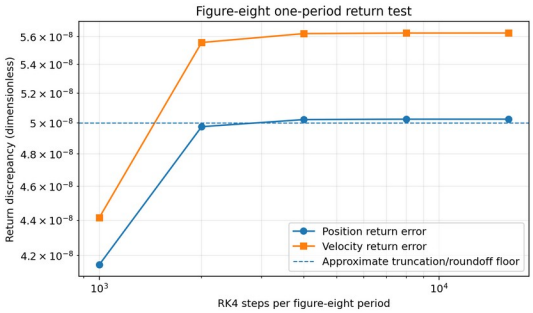


Figure 20. Figure-eight one-period return discrepancy for RK4. The plateau reflects rounded initial conditions and period, limiting its use as a convergence reference.

The equal-mass figure-eight choreography was integrated with RK4 for exactly one nominal period $T \approx 6.32591398$ using the rounded initial conditions of Chenciner and Montgomery (2000). The position return discrepancy decreases from 4.15×10^{-8} at 1,000 steps per period, then plateaus near 5.03×10^{-8}; the velocity return discrepancy similarly plateaus near 5.63×10^{-8}. This floor is consistent with the finite precision of the published initial conditions and period, so the test is reported as a return check, not as evidence of fourth-order convergence. The Pythagorean problem, originating with Burrau (1913) and later studied by Szebehely and Peters (1967), is under-resolved in this implementation and is not used to rank integrators.

6. Discussion

The long-duration eccentric experiment directly tests the paper's central distinction. At 10,000 steps per orbit over 1,000 periods, velocity-Verlet and Yoshida4 maintained bounded energy-error envelopes, but their phase errors still accumulated. RK4 delivered lower phase error than velocity-Verlet at the final time for this particular step size, while its energy error showed a secular trend. Yoshida4 had the smallest measured phase discrepancy among the fixed-step methods in this setup. These results do not imply that symplectic methods always give more accurate trajectories; rather, they show why conservation and pointwise accuracy must be reported separately.

The adaptive DOP853 baseline is essential context: at 2,975 function evaluations it reached a maximum discrepancy of 1.11×10^{-11}a over the one-orbit eccentric test, much smaller than the fixed-step points shown in the work–precision plot. It would be misleading to describe RK4 as highly accurate per evaluation without acknowledging that result. The work–precision comparison is a finite-interval benchmark; long-duration adaptive behaviour was not measured here.

The observed circular-orbit energy slope of 3.32 versus 2.04 for $e = 0.9$ is consistent with a problem-specific cancellation hypothesis, but does not prove it. The finest circular points are influenced by the floating-point floor. Likewise, the figure-eight return plateau is limited by rounded initial conditions, and the Pythagorean close

encounter requires a better-resolved, independently checked integration. The direct-force scaling trend is consistent with $O(N^2)$, but the short timing kernel does not establish a hardware-independent performance law.

7. Limitations

The model is Newtonian and excludes relativistic corrections, solar oblateness, radiation pressure, tides, dissipation, and precision planetary ephemerides. The long eccentric run uses one fixed timestep and one orbital configuration; other eccentricities and timestep choices could change the balance between energy and phase error. No direct comparison against REBOUND, WHFast, IAS15, or JPL Horizons was completed. Consequently, this manuscript does not claim external validation, predictive accuracy for real Solar System ephemerides, or superiority over established N-body packages. The Pythagorean integration remains under-resolved and is described only as a cautionary case.

8. Reproducibility, data availability, and AI-use disclosure

The source code, supporting numerical outputs, figures, and reproducibility materials for this study are available in the public GitHub repository: https://github.com/DurodolaDaniel/nbody-gravitational-integration-benchmark. The repository contains the direct-force implementation, convergence and work–precision scripts, CSV outputs, figures, and environment metadata. Array operations use NumPy (Harris et al., 2020), adaptive integration uses SciPy (Virtanen et al., 2020), and compiled benchmark kernels use Numba (Lam et al., 2015). The computational environment recorded Python 3.13.5, NumPy 2.3.5, SciPy 1.17.0, Matplotlib 3.10.8, Numba 0.65.1, and pandas 2.2.3 on Linux x86_64 with an AMD EPYC 9V74 80-Core Processor visible to the runtime (three logical CPUs available to the process). Run commands and initial conditions are documented in the repository. A persistent archive DOI will be added after the corrected release has been deposited in Zenodo.

AI-use disclosure: Generative AI tools assisted with code review, manuscript editing, and identifying numerical checks. The author is responsible for verifying the equations, implementation, measurements, references, and claims in the final manuscript.

9. Conclusion

This study benchmarks Euler, velocity-Verlet, RK4, and Yoshida4 in direct Newtonian gravitational integration using analytical two-body references, convergence tests, invariant diagnostics, long-duration eccentric motion, perturbation and binary tests, scaling, and softening sensitivity. Fine-grid tests recover first-order Euler, second-order velocity-Verlet, and fourth-order RK4 and Yoshida4 position convergence where the fits remain above the roundoff floor. Over 1,000 eccentric periods, the symplectic methods retain bounded energy error while phase error grows, demonstrating why conservation alone cannot guarantee trajectory fidelity. The adaptive DOP853 baseline substantially outperforms the fixed-step points in the one-orbit work–precision test. The contribution is a transparent benchmark and numerical interpretation, not a new algorithm. The source code and supporting materials are available in the public repository; a persistent archive DOI will be added after the corrected release is deposited in Zenodo. External REBOUND/JPL Horizons validation remains future work.

References

Aarseth, S. J. (2003). Gravitational N-Body Simulations: Tools and Algorithms. Cambridge University Press. https://doi.org/10.1017/CBO9780511535246.

Barnes, J., & Hut, P. (1986). A hierarchical O(N log N) force-calculation algorithm. Nature, 324, 446–449. https://doi.org/10.1038/324446a0.

Brouwer, D. (1937). On the accumulation of errors in numerical integration. The Astronomical Journal, 46, 149–153. https://doi.org/10.1086/105361.

Burrau, C. (1913). Numerische Berechnung eines Spezialfalles des Dreikörperproblems. Astronomische Nachrichten, 195, 113–118. https://doi.org/10.1002/asna.19131950602.

Chenciner, A., & Montgomery, R. (2000). A remarkable periodic solution of the three-body problem in the case of equal masses. Annals of Mathematics, 152(3), 881–901. https://doi.org/10.2307/2661357.

Hairer, E., Lubich, C., & Wanner, G. (2006). Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations (2nd ed.). Springer. https://doi.org/10.1007/3-540-30666-8.

Hairer, E., Nørsett, S. P., & Wanner, G. (1993). Solving Ordinary Differential Equations I: Nonstiff Problems (2nd revised ed.). Springer. https://doi.org/10.1007/978-3-540-78862-1.

Harfst, S., Gualandris, A., Merritt, D., Spurzem, R., Portegies Zwart, S., & Berczik, P. (2007). Performance analysis of direct N-body algorithms on special-purpose supercomputers. New Astronomy, 12(5), 357–377. https://doi.org/10.1016/j.newast.2006.11.003.

Harris, C. R., Millman, K. J., van der Walt, S. J., et al. (2020). Array programming with NumPy. Nature, 585, 357–362. https://doi.org/10.1038/s41586-020-2649-2.

International Astronomical Union. (2012). Resolution B2: Re-definition of the astronomical unit of length. XXVIIIth General Assembly. https://www.iau.org/static/resolutions/IAU2012_English.pdf.

Lam, S. K., Pitrou, A., & Seibert, S. (2015). Numba: A LLVM-based Python JIT compiler. In Proceedings of the Second Workshop on the LLVM Compiler Infrastructure in HPC. https://doi.org/10.1145/2833157.2833162.

Rein, H., & Liu, S.-F. (2012). REBOUND: An open-source multi-purpose N-body code for collisional dynamics. Astronomy & Astrophysics, 537, A128. https://doi.org/10.1051/0004-6361/201118085.

Rein, H., & Spiegel, D. S. (2015). IAS15: A fast, adaptive, high-order integrator for gravitational dynamics, accurate to machine precision over a billion orbits. Monthly Notices of the Royal Astronomical Society, 446(2), 1424–1437. https://doi.org/10.1093/mnras/stu2164.

Rein, H., & Tamayo, D. (2015). WHFast: A fast and unbiased implementation of a symplectic Wisdom–Holman integrator for long-term gravitational simulations. Monthly Notices of the Royal Astronomical Society, 452(1), 376–388. https://doi.org/10.1093/mnras/stv125.

Szebehely, V., & Peters, C. F. (1967). Complete solution of a general problem of three bodies. The Astronomical Journal, 72, 876. https://doi.org/10.1086/110355.

Virtanen, P., Gommers, R., Oliphant, T. E., et al. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. Nature Methods, 17, 261–272. https://doi.org/10.1038/s41592-019-0686-2.

Wisdom, J., & Holman, M. (1991). Symplectic maps for the N-body problem. The Astronomical Journal, 102, 1528–1538. https://doi.org/10.1086/115978.

Yoshida, H. (1990). Construction of higher order symplectic integrators. Physics Letters A, 150(5–7), 262–268. https://doi.org/10.1016/0375-9601(90)90092-3.